

FloodVis: Deep Learning with Aerial Imagery for Flood Assessment

Project Team

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Session 2021-2025

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June, 2025

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Chapter 1

Introduction

This document outlines the comprehensive development of the FloodVis system, detailing its problem statement, motivation, and proposed solution. FloodVis aims to enhance flood disaster response and management using deep learning models, satellite, and UAV imagery for flood detection and survivor identification. The document serves developers, project managers, marketing staff, testers, users, and documentation writers by providing detailed descriptions of the system's requirements, use cases, functional and non-functional requirements, architecture, and design models.

1.1 Problem Statement

The 2022 floods in Pakistan left 1,739 people dead and more than 8 million displaced. Floods are among the most catastrophic natural disasters, causing significant loss of life and extensive damage to land and critical infrastructure. The scale of these disasters often overwhelms existing response systems, delaying the delivery of vital data to decision-makers and slowing down rescue efforts.

Traditionally, ground-based and satellite remote sensing equipment has been the go-to solution for flood monitoring and damage assessment. Optical satellite images provide rich visual details but are hindered by cloud cover and other atmospheric conditions, which can delay accurate flood mapping.

Moreover, the identification of survivors and animals in flood-affected regions is typically a time-consuming and labor-intensive process, adding further delays to rescue operations. The need for a more efficient and automated solution that integrates SAR satellite imagery and UAV imagery with deep learning models is clear, as it could provide real-time data and improve the accuracy and speed of flood detection, survivor identification, and damage assessment.

1.2 Scope

The scope of this project includes the development and enhancement of flood monitoring and detection through a flood disaster management system that integrates deep learning with SAR satellite and UAV imagery. The system will perform critical functions such as flood detection, mapping affected areas, and detecting humans and animals in flood zones. To achieve this, FloodVis will acquire, analyze, and interpret satellite images and UAV aerial images. Flood exposure zones will be mapped using semantic segmentation techniques.

The system will use custom object detection models to recognize and count affected individuals, livestock, and damaged buildings. It will provide approximate coordinates for rescue teams, enabling faster and more accurate deployment of resources. FloodVis will consist of a backend system that focuses on data storing, model deployment and APIs, while the front-end will contain a dashboard with information and reports on the effects of floods. The models will also be deployed on the cloud, allowing scalability especially during large scale disasters.

Project boundaries include: detecting floods and providing data analytics (exclude handling ground-based rescue operations or non-flood-related disasters). This solution will focus solely on improving response times and resource allocation during large-scale floods for disaster relief teams, making it a critical tool for disaster management in vulnerable regions.

1.3 Modules

1.3.1 Module 1 (Data Collection + Preprocessing)

1. Acquire, clean, augment and generate mask labels for optical satellite imagery of flooded regions (pre, during, post-flood) from Sentinel-1 and Sentinel-2 satellites through Google Earth Engine and/or Sentinel Hub.
2. Acquire, clean, augment and label aerial imagery of floods having visible humans, animals and infrastructure from the internet and/or a simulated environment.

1.4 User Classes and Characteristics

User class	Description
Emergency Responder	Emergency responders are individuals who partake in, as well as coordinate, rescue and relief efforts during flood events. Emergency responders will use FloodVis to acquire the count and locations of humans and animals stranded in the flood. They also need to stay up-to-date on the flood's progression. All this information can be obtained through the web application before the responders head out into the field. They are individuals involved on-field in more hands-on activities such as evacuation and providing aid. They are collaborative and resourceful individuals who can assess the severity of a situation and take quick actions.
Disaster Manager	Disaster managers are people who make the key decisions regarding resource allocation during and after the flood. They also keep a check on the general state of the flood response. Disaster managers may also use the web application in order to prepare for a flood event. They are individuals with strong analytical skills. They are fast decision-makers who ensure effective communication.
System Administrator	A system administrator is responsible for maintaining the FloodVis application, ensuring it runs smoothly and securely. They ensure that users are properly authenticated, that the system remains fully operational, that data integrity is preserved, and that security breaches are prevented. Additionally, they implement regular patches and updates to keep the application up-to-date and secure. System administrators are IT experts who have a strong grasp on computer science concepts such as Information Security, Computer Networks, and Database Systems. They can quickly diagnose and resolve technical issues in the system.

Chapter 2

Project Requirements

2.1 Use-cases

2.1.1 Use Case Diagram

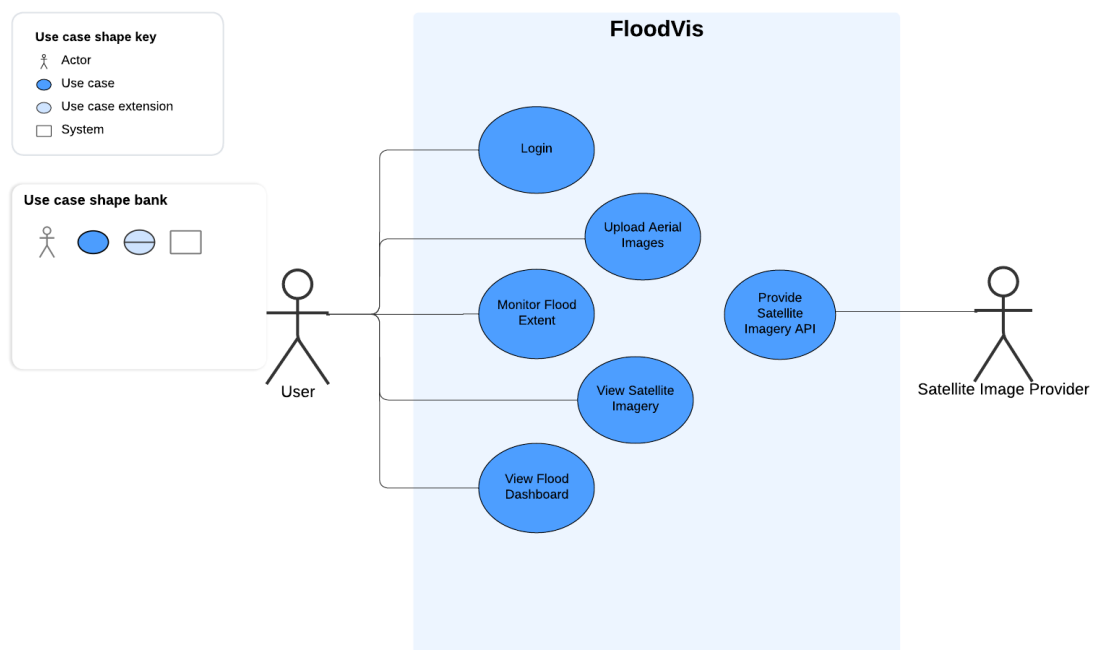


Figure 2.1: Use Case Diagram

2.1.2 High Level Use Case

Table 2.1: Use Case 1

UC01	Provide Satellite Imagery API
Actors	User, External System
Type	Primary
Description	In this use case, the user will be able to connect to the external satellite imagery provider through an API. The user can then request satellite imagery of a specified area of interest.

2.1.3 Extended Use Case

Table 2.2: Extended Use Case 1

Use Case	Provide Satellite Imagery API
Scope	FloodVis Web Application
Level	User goal
Primary Actor	External System
Stakeholder(s) and Interests	• User: Wants access to current and accurate satellite imagery for monitoring flood risks. • System Administrator: Needs to guarantee reliable integration of external APIs for real-time image retrieval. • Rescue Teams: Depend on up-to-date satellite imagery to assess potential risks in flood-prone areas.
Pre-condition(s)	• The system has an active connection to the external satellite imagery provider's API. • The satellite imagery provider is operational.
Post-condition(s)	The system successfully retrieves and stores the satellite imagery.

Main Success Scenario	User	External System
	1. The user opens the application, which automatically connects to the satellite imagery provider by sending an API key.	
		2. The satellite imagery provider authenticates the API key to establish a connection.
	3. The user can request satellite imagery for a specified area on a specified date.	
		4. The provider checks if images of the specified area on the date are available.
		5. If the imagery is available then the provider returns it, otherwise it returns the latest imagery available of the area.
Extensions	• If the satellite provider's response is delayed, the system tries again after a set time. • If the requested region's imagery is unavailable, the system retrieves the latest available imagery and notifies the user.	
Special Requirements	• The API connection must be reliable to ensure timely satellite imagery retrieval. • The system must handle large satellite data efficiently.	

Technology and Data Variations	• The format of the satellite images (resolution, file type) may vary depending on the provider. • Different APIs may offer varying levels of data (e.g., historical imagery vs. real-time imagery).
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2.2 Functional Requirements

1. The system should classify regions as flooded or non-flooded.
2. The system should detect and count the number of critically affected and non-critically non-affected humans.
3. The system should detect and count the number of critically affected and non-critically affected cattle.
4. The system should detect number of submerged infrastructure.
5. The system should provide quantitative data via a dashboard, including flood area size and severity of the flood.
6. The system should allow users to export reports containing flood analysis and detection summaries.
7. The system should allow the comparison of pre- and post-flood images to assess the extent of damage and the speed of water receding.

2.3 Non-Functional Requirements

1. The system should be able to process images with varying levels of noise.
2. The system should support medium-resolution satellite and UAV images for flood assessment.
3. The system should be able to process satellite and UAV images captured from different angles for better flood mapping.
4. The system must provide accurate flood segmentation and detection of cattle, humans, and affected infrastructure.
5. The system should process large amounts of data (satellite and UAV images) quickly to provide insights during emergencies.
6. The interface should be user-friendly, enabling decision-makers and rescue teams to quickly understand the information presented.
7. The system should ensure the confidentiality and integrity of satellite and UAV data, as well as the processed results, especially in sensitive disaster scenarios.

2.4 Domain Model

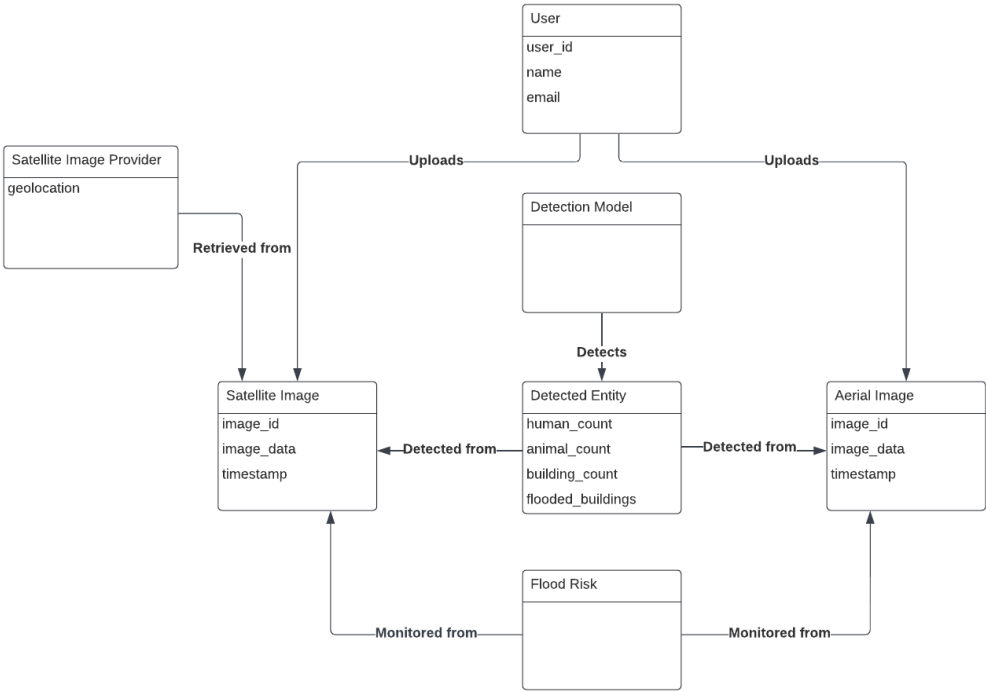


Figure 2.2: Domain Model

Chapter 3

System Overview

3.1 Architectural Design

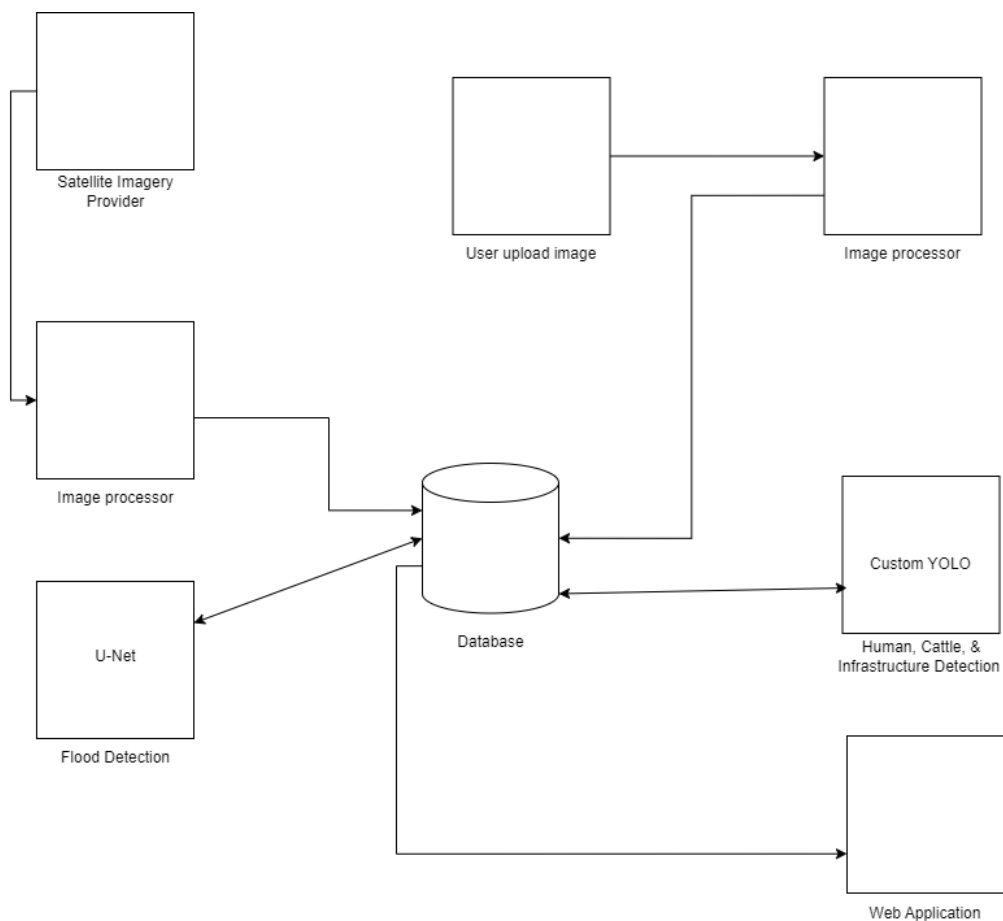


Figure 3.1: Box-line Diagram

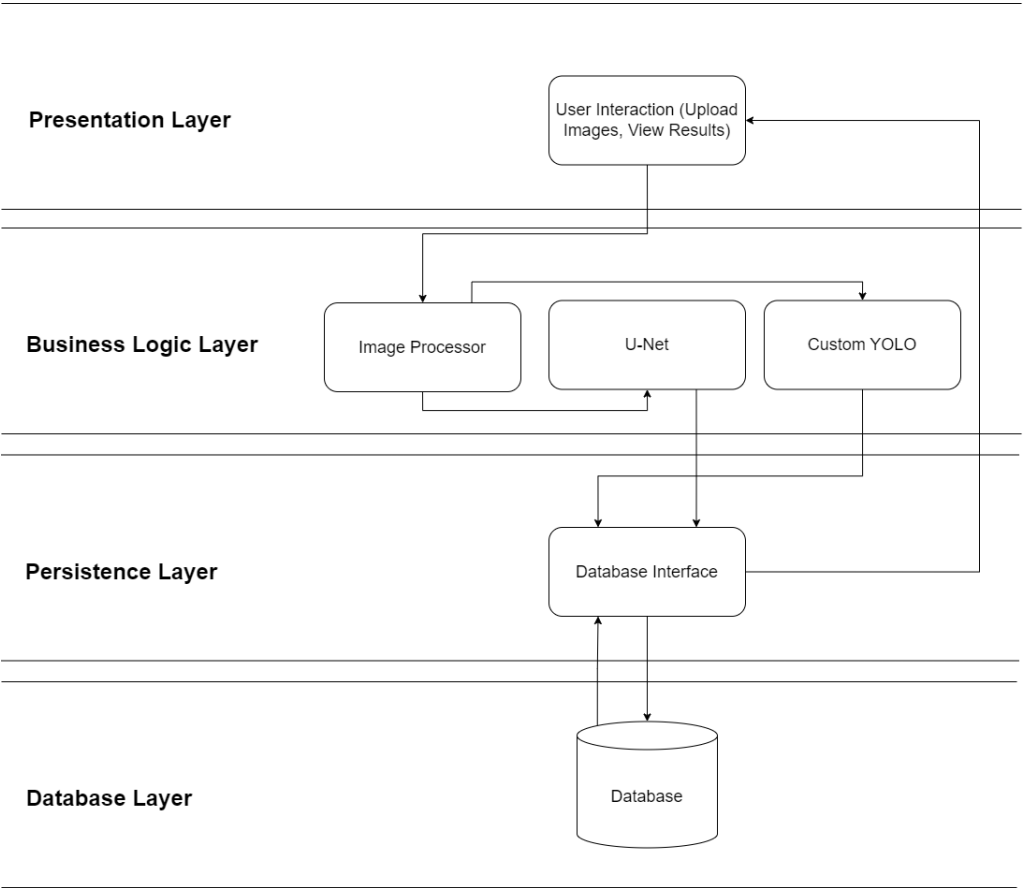


Figure 3.2: Layered Architecture Diagram

3.2 Design Models

3.2.1 Activity Diagram

Activity Diagram

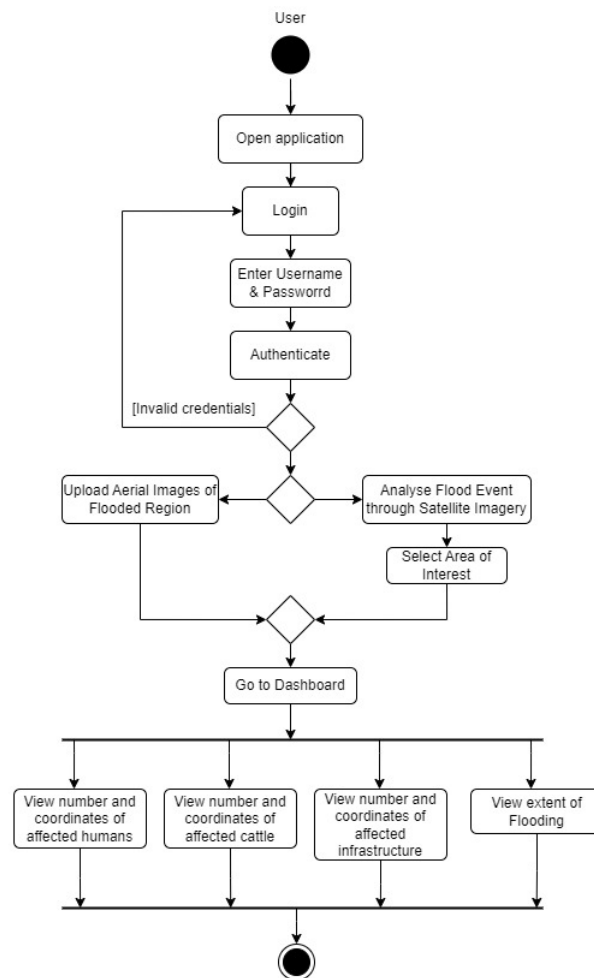


Figure 3.3: Activity Diagram

3.2.2 State Transition Diagram

State Diagram

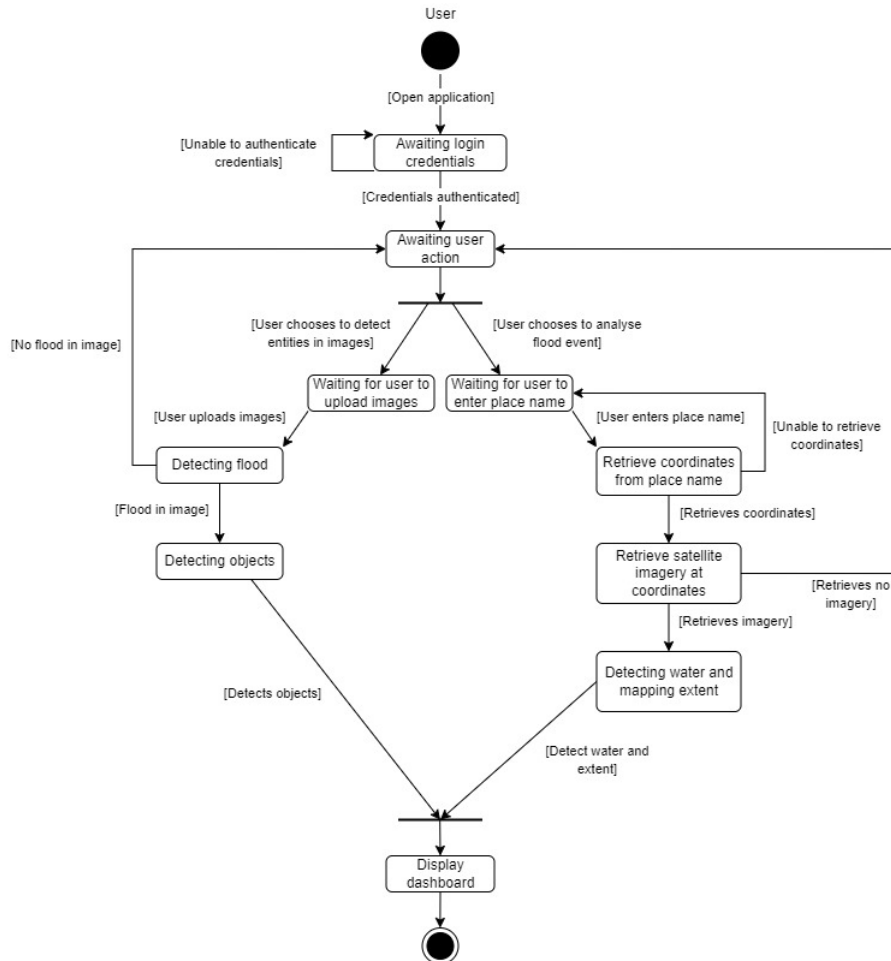


Figure 3.4: State Transition Diagram

3.2.3 Class Diagram



Figure 3.5: Class Diagram

3.2.4 Data Flow Diagram

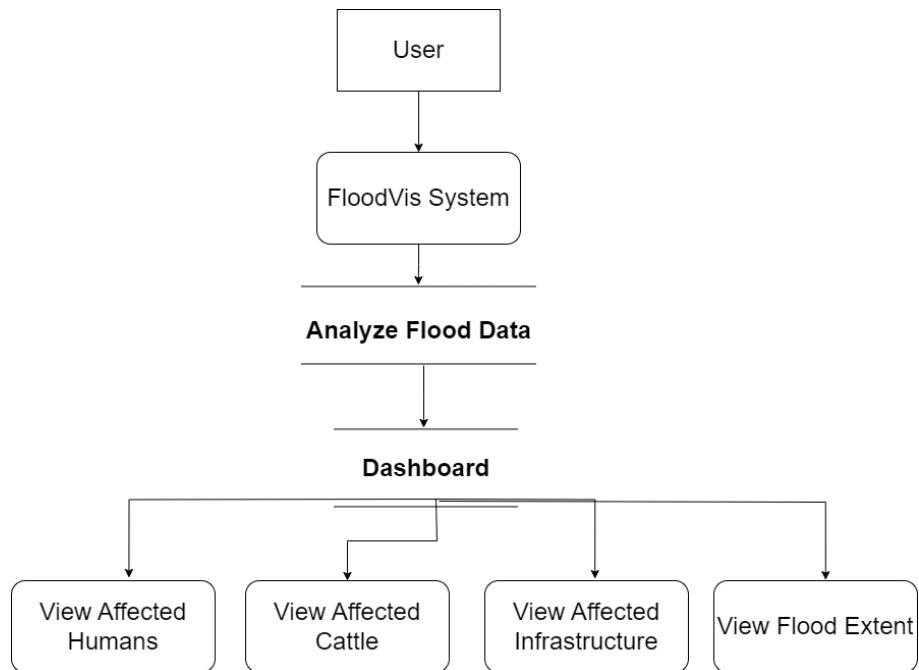


Figure 3.6: Data Flow Diagram Level 0

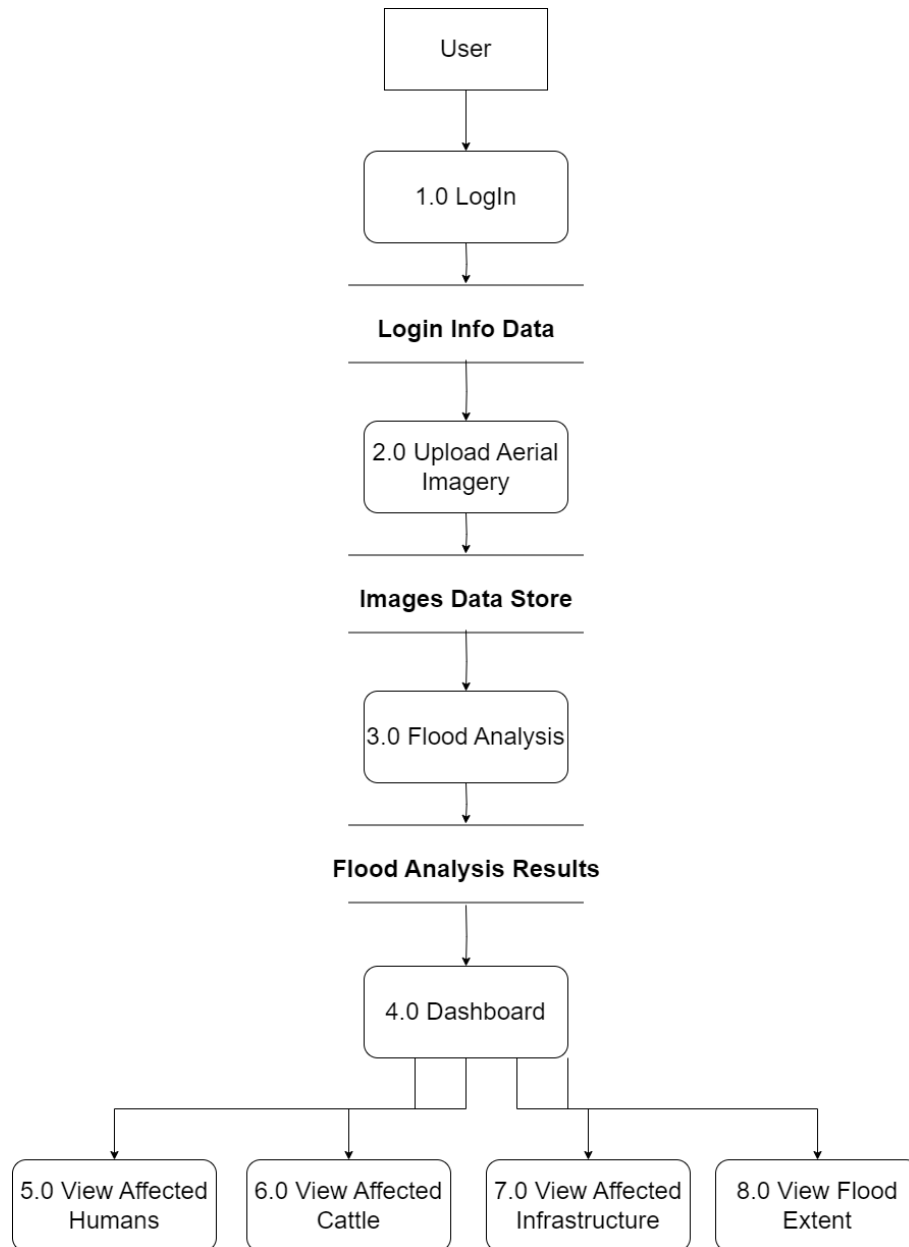


Figure 3.7: Data Flow Diagram Level 1



Figure 3.8: Data Flow Diagram Level 2

3.2.5 Sequence Diagram

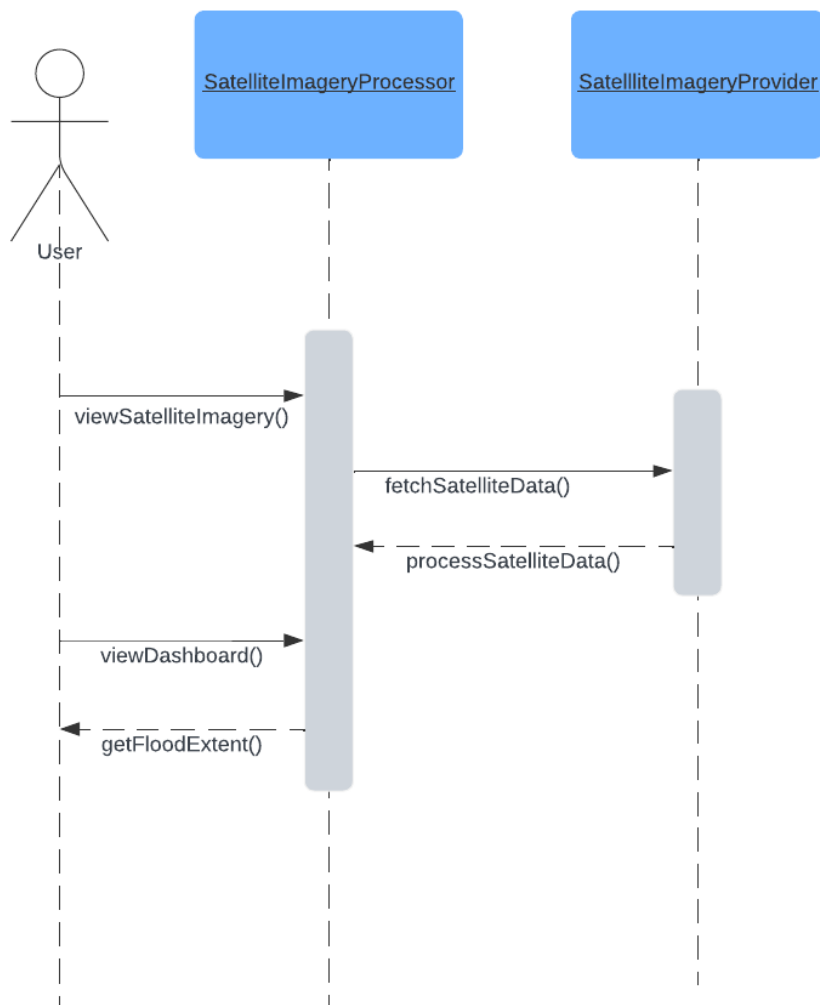


Figure 3.9: SD

3.2.6 State Sequence Diagram

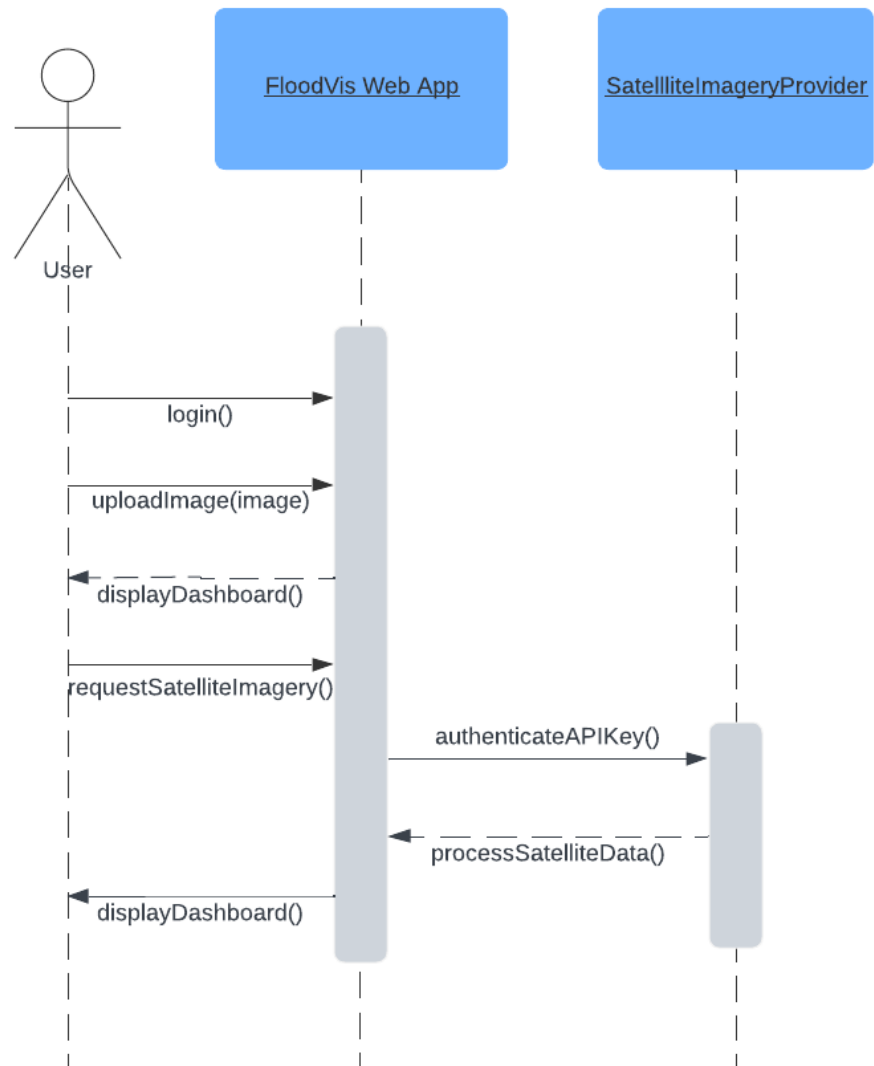


Figure 3.10: SSD

3.3 Data Design

3.3.1 ERD

ERD

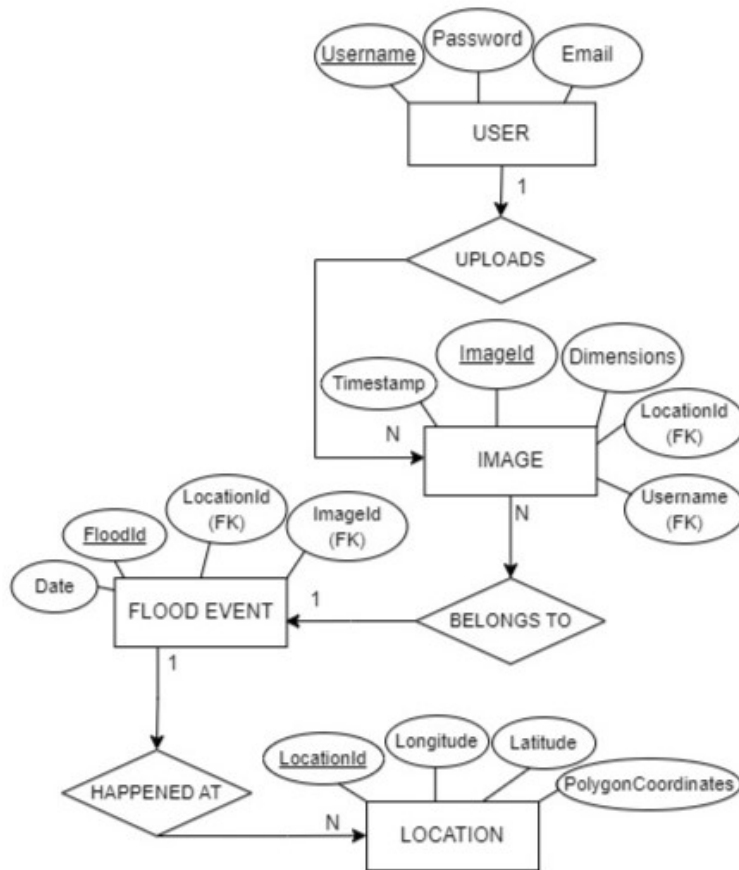


Figure 3.11: ERD

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